Mustafa Ebrahim Chasmai | **2019CS10341**
Computer Science and Engineering | **B.Tech.**
Indian Institute of Technology Delhi | **DOB: 22-10-2001**

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EDUCATION

Examination	Body	Institute	Year	CGPA/%
First Year	IIT Delhi	IIT Delhi	2020	9.73
Intermediate/2+	Maharashtra state board	S.P. College, Pune	2019	89.08
Matriculation	Maharashtra state board	Rosary School, Pune	2017	93.04

ACADEMIC ACHIEVEMENTS AND SCHOLARSHIPS

- Secured **All India Rank 91** in IIT JEE Advanced 2019 (out of more than 1,50,000 candidates). *(2019)*
- Awarded **top 7% merit** award **twice**, on academic merit, out of 1000+ students, during the first year. *(2019)*
- Awarded Gold Medal and Certificate in **Indian National Physics Olympiad (INPhO)**. *(2019)*
- Qualified for the **International Physics Olympiad** camp (OCSC), being among the top 35 students in the country to do so and the only one from Pune. *(2019)*
- Awarded for being amongst the **National top 1%** in the **National Standard Examination** in Physics **(NSEP)** in 11th & 12th standards and in Chemistry **(NSEC)** in 12th standard. *(2018, 19)*
- Conferred the **KVPY** Fellowship by the Department of Science and Technology, Government of India with an **All India Rank of 209.** *(2018)*
- Awarded Scholarship for higher education (SHI) under Innovation in Science Pursuit for Inspired Research **(INSPIRE)** by the Ministry of Science & Technology, Govt. of India. *(2017)*
- Secured 16+ medals and certificates in Science, Maths and English **International Olympiads** conducted by **SOF and Silverzone**. *(2014 - 19)*
- Qualified for the **Maharashtra State Scholarship** in 4th and 7th standard. *(2011, 14)*

INTERNSHIPS

Farm Boundary Detection — **Satsure, Bangalore**
Data Science Intern | Deep Learning | Computer Vision | Remote Sensing — *Dec 2020*

- Worked on obtaining clear and accurate boundaries for Indian farms. Used **Sentinel 2 satellite data**.
- Built a complete pipeline for extracting vector shape files directly from raster files using a **Mask-RCNN model** for **instance segmentation**.
- Explored basics of **remote sensing**. Used multiple wavelength information to get better results. Worked on obtaining NDVI indices and other post-processing products.

Automated Box Counting — **Flytbase, Pune**
AI and Robotics Intern | Deep Learning | Computer Vision — *Apr 2020 - July 2020*

- Leveraged various **Deep Learning** and **Computer Vision** techniques for the unsolved problem of automated box localisation and counting in Bulk storage warehouses for the **drone-based** automated inventory management system. Starting from scratch, completed the project and achieved acceptable accuracy.
- Implemented **Object Detection, Instance Segmentation,** Unsupervised Intrinsic Decomposition, **Corner Detection, Edge Detection**, and many other approaches.
- Experimented with SSD, Retinanet, Mask-RCNN, KittiSeg, DexiNed and VGG and also a custom model architecture from ground-up.
- Used Flask to develop a **web app** for encapsulating the project with an intuitive interface that facilitated manual intervention and automated annotations, leading to an **automated Online Machine Learning** system, enabling generation of new trainable data (with annotations) as the project is being used

Automatic Stock Replenishment App — **ShoppinPal, Pune**
Software Developer | Web Development | UI | Angular — *Dec 2019*

- Worked on front-end web development for **StockUp (automatic stock replenishment app).**
- Learned basics of Angular, NodeJS and MongoDb and general backend development.
- Deployed a remote setup and touched basics of DevOps.

PROJECTS

Authentication in Automotive Networks | Research Project **IIT Delhi**
Prof. Vireshwar Kumar, CSE IITD | Network Security | Controller Area Network *Nov 2020 - present*

- Exploring **new vulnerabilities in CAN** which can be exploited **remotely** to gain control of the vehicle.
- Developing novel mechanisms which can be readily implemented in **resource-constrained** CAN.
- Working on **Intrusion Detection** techniques and **Trusted Execution Environments** (TEEs).

Dynamic Memory Allocation | Course Project **IIT Delhi**
Prof. Rahul Garg, CSE IITD | Course: Data Structures & Algorithms *Sep 2020 - Dec 2020*

- Built a java-based simulated dynamic memory allocator, with **allocate, free** and **de-fragment** operations, modelling different processes involved in the actual allocator.
- Optimized efficiency of operations using the abstract **data structures** of **AVL Trees**.

Digital Image Filtering | Course Project **IIT Delhi**
Prof. Anshul Kumar, CSE IITD | Course: Digital Logic & System Design *Sep 2020 - Dec 2020*

- Designed a system to apply **smoothening** and **sharpening** filters on an FPGA using VHDL.
- Used a 3 x 3 sliding window for applying filtering operation on QQVGA images.

KEY COURSES UNDERTAKEN

MTL145*:	Number Theory	Congruences, RSA, Euler's Thm, Diophantine Approximations
MTL106:	Probability & Stochastic Processes	Sigma algebra, Random Variables, Markov Chains & Processes
COL106:	Data Structures and Algorithms	Hashing, Trees, Kruskal's & Prim's algorithm, Stable Marriage
COL202:	Discrete Mathematical Structures	Boolean Logic, Graph Theory, Cuts, k-Spanners, Sets, Lattices

(* - Ongoing course)

KEY ONLINE COURSES UNDERTAKEN

Deep Learning	Fundamentals of Deep Learning and AI.	*(Coursera)*
Adv. Machine Learning	Advanced concepts and engineering practices in AI and ML	*(Coursera)*
Self-driving Cars	Static and Dynamic Localisation, Mapping, Planning and Control	*(Coursera)*
Natural Language Processing	Markov models, word encodings, LSTMs and attention models	*(Coursera)*
Complexity Analysis	Dynamic systems, Chaos theory, Fractals	(Santa Fe institute)

INTERESTS

Machine Learning	Self-Supervised learning, Clustering, Probabilistic models, Reinforcement Learning
Deep Learning	Generative networks, GANs, Siamese Networks, Single Shot Detectors
Computer Vision	Stitching, Pattern Recognition, Object Detection, Segmentation
Web Development	Frontend Web Development and designing, Backend Data management
Cosmology	Relativity, the Multiverse Theory, Black Holes, The Big Bang, String Theory

POSITIONS OF RESPONSIBILITY

Founding Member | AI & ML Club IIT Delhi (2020)

- Organized **podcasts** by professionals and professors in the fields of AI and ML, created **tutorials** for beginners and set up **discussion forums** to build a community of **AI & ML enthusiasts** at IITD.
- Head of the **Designing** team and involved in building and showcasing different **projects** to attract students.

Academic Mentor | COL100: Introduction to Computer Science | Board of Student Welfare, IIT Delhi (2020)

- Responsible for providing **academic help and guidance** to 40+ freshmen in the course.

Team-Head | Tryst 2020 | Technical festival of IIT Delhi | Delayed due to COVID-19 (2020)

- Managed a team responsible for the publicity of various events of Tryst in IITD and other colleges.

Senior House Prefect | Rosary High School, Pune (2014)

- Led 200+ students from all divisions, in conceptualizing and executing school sports and cultural events.

EXTRA CURRICULAR ACTIVITIES

- Enthusiastic **Rubik's Cube** solver (solved physical upto **9 x 9 x 9**, virtual upto **20 x 20 x 20**)
- Undergoing training in **athletics** under the National Sports organisation (**NSO**).
- Won various medals in **Swimming** across different annual SLTC Swimming Competitions organised by the Rashtriya Life Saving Society (**RLSS**).
- Actively participated in many **photography and photo editing** competitions in the **PFC club**, IITD.